Project Results & Calculated Inventories

1. Herbaceous Layer Biomass Conversions

1.1 Processed Biomass Yields

- Sample 10: 43.32 g/0.25m² → 173.28 g/m² → 1.7328 tons/ha
- Sample 20: 73.76 g/0.25m² → 295.04 g/m² → 2.9504 tons/ha
- Sample 30: 51.12 g/0.25m² → 204.48 g/m² → 2.0448 tons/ha
- Sample 40: 38.65 g/0.25m² → 154.60 g/m² → 1.5460 tons/ha
- Biomass Statistical Summary: Mean Yield = 2.0685 tons/ha | Standard Deviation = 0.6229 tons/ha

Table 1.1: Herbaceous Community Composition & Abundance Hierarchy (Dry Weight Rank)

Species Name	Species Code	Count Rank 1	Count Rank 2	Count Rank 3	Frequency	DWR Abundance (%)
Eragrostis rigidior	Erarig	23	4	0	27	63.0%
Schizachyrium sanguineum	red leaves	0	18	2	20	20.0%
Brachiaria brizantha	hairy leaf	0	2	26	28	10.1%
Hygrostis scabra	SNO	4	0	1	5	4.2%
Semi-digitate inflorescence	semi	2	0	0	2	2.1%
Aristida congesta	ARI	1	1	0	2	0.4%
Elongated digitate inflores.	Eldigfl	0	1	0	1	0.2%
Gross 2	gross2	0	1	0	1	0.0%
Hetcon	het con	0	1	0	1	0.0%
Speargrass	spear grass	0	1	0	1	0.0%
Poly	poly	0	1	0	1	0.0%
Posurv	Pos urv	0	0	1	1	0.0%
Fabacae unkj	pod	0	0	0	0	0.0%
TOTAL		**30**	**30**	**30**	**90**	**100.0%**

2. Savanna Woody Stand Architectures

- Combined Canopy Coverage Footprint: Summed area = 4.17 m²
- Stand Level Percent Canopy Cover: 5.2% (Indicates a classic open wood-savanna system)
- Canopy Strata Heights: Mean Height = 1.84 m | Maximum Canopy Height = 2.25 m

Table 2.1: Taxonomic Biomass Stock Summaries (Pivot Table Output)

Tree Genus / Species Classification	Number of Individuals	Total Harvested Weight (kg)	Aboveground Biomass (tons/ha)
Acacia caffra	2	0.00080	0.00010

Acacia caffra / Combretum apiculatum blend | 1 | 0.00082 | 0.00010
Combretum apiculatum | 3 | 0.00858 | 0.00107
Dichrostachys cinerea | 1 | 0.00262 | 0.00033
NET TRANSECT STAND TOTALS | 7 | 0.01282 kg | 0.00160 tons/ha

3. Physical & Hydrological Soil Profiles

- Calculated Water Displacement Core Volume (V): 35.03 ml
- Core Oven-Dry Net Mass: 74.99 g (Corrected for moisture weight loss)
- Final Profile Bulk Density (ρ): 2.14 Mg m^{-3}
- Calculated Soil Porosity Fraction (θ_sat): 1.89%
- Whole Soil Base Sample Dry Weight: 2142.73 g
- Gravel Mass Fraction Percentage: 26.9% | Gravel Volume Fraction Space Occupancy: 21.8%
- Corrected Non-Gravel Fine Fraction Bulk Density (ρ_fines): 2.00 Mg m^{-3}
- Textural Grading Values: Clay = 20.19% | Silt = 20.26% | Sand = 59.55%
- Final Mechanical Textural Class: Sandy Clay

Table 3.1: Hydrological Potentials & Multi-Horizon Storage Budgets
Soil Profile Layer	Horizon Depth Range	Layer Thickness (z)	Water Holding Capacity (mm)	Plant Available Water Capacity (mm)
Layer 1	0 to 390 mm	0.39 m	0.131 mm	0.117 mm
Layer 2	390 to 650 mm	0.26 m	0.167 mm	0.142 mm
Layer 3	650 to 920 mm	0.27 m	0.265 mm	0.249 mm
**INTEGRATED PROFILE TOTALS	0 to 920 mm	0.92 m	0.563 mm	0.508 mm**

4. Soil Hydrochemistry & Budget Models

- Temperature-Corrected Conductivity (EC_corrected): 16.74 mS m^{-1}
- Net Extracted Exchangeable Acidity: 0.80 cmol(H^+) kg^{-1}

Table 4.1: Chemical Sorption, Exchange Capacities, and Base Saturation
Displaced Cation Group	Measured Structural Concentration (cmol(+) kg^{-1})	Percentage Contribution to ECEC (%)
Calcium (Ca^{2+})	1.307	56.2%
Magnesium (Mg^{2+})	0.777	33.4%
Potassium (K^+)	0.200	8.6%
Sodium (Na^+)	0.043	1.8%
Exchangeable Acidity (Al^{3+}/H^+)	0.800	34.4% *(Exceeds base balance ratio)*
**Effective CEC (ECEC)	2.327 cmol(+) kg^{-1}	Net Base Saturation (%BS) = 79.5%**

Table 4.2: Landscape Carbon & Nitrogen Storage Inventories

Ecosystem Pool Component | Carbon Density Stock (g C/m²) | Nitrogen Density Stock (g N/m²) | Stoichiometric Ratio (C:N)

Topsoil Soil Matrix (0-1m) | 1450.40 | 112.50 | 12.89 : 1

Living Savanna Tree Biomass | 14.20 | 2.10 | 6.76 : 1

COMBINED SYSTEM INVENTORY | 1464.60 g C/m² | 114.60 g N/m² | 12.78 : 1

4.3 Nitrogen Transformation Rates & Productivity Potentials

- Net Inorganic Nitrogen Change (ΔN): 7.00 mg N/L
- Anaerobic Nitrogen Mineralisation Index (N_min): 9.80 mg N / ml soil / day
- Annualized Topsoil Nutrient Models: Turnover Rate = 27.78% yr^{-1}
- Net Primary Productivity Support Potential (NPP): 500 g DM yr^{-1}

5. Eco-Topographic Catena Co-Variations

Table 5.1: Landscape Catchment Co-Variation Matrix

Slope Catchment Zone | Sand (%) | Clay (%) | Grass Yield (tons/ha) | Woody Mass (kg) | Mean Tree Height (m) | Max Tree Height (m)

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Top Ridge | 69.4% | 12.9% | 1.36 | 0.07 | 1.80 | 3.47

Middle Slope | 66.0% | 17.0% | 1.54 | 0.06 | 1.92 | 2.95

Bottom Slope | 36.0% | 31.6% | 2.54 | 0.05 | 1.51 | 1.93

6. Aquatic Channel Hydrodynamics

- Total Integrated River Discharge (D_total): 0.973 $m^3\ s^{-1}$ (cumecs)
- Physico-Chemical Baseline Limits: Saturated Oxygen = 7.39 mg/L (78.6%) | Alkaline pH = 8.65 | High EC = 379.8 µS/cm